IROS 2026 · AIM–Ctrl

ROBUST TO WHAT?

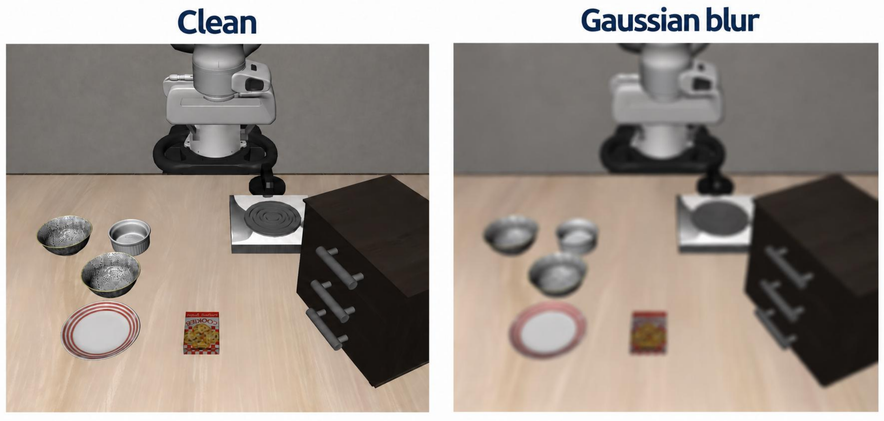
Interpreting Shifted Success in Executable Vision–Language–Action Policies

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What does a success rate hide?

The total hides gains and losses



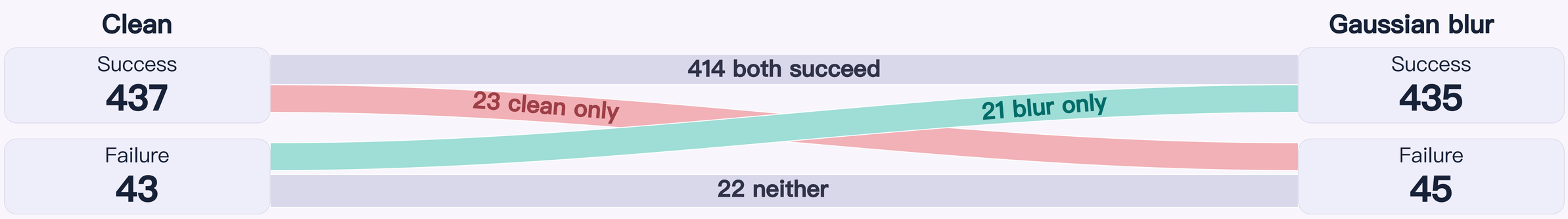
Gaussian blur (σ = 1 px); image exaggerated.

MATCH ONE CASE BY

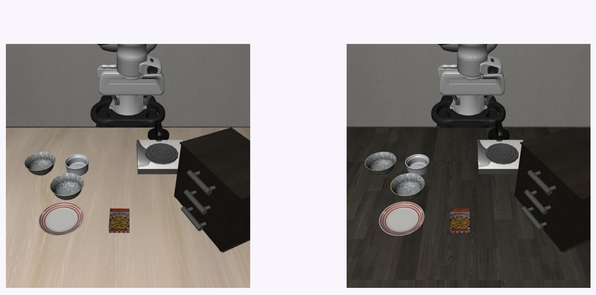
π_0	LIBERO	Official
checkpoint	**task**	**initial–state index**

Clean run | **Blurred run**

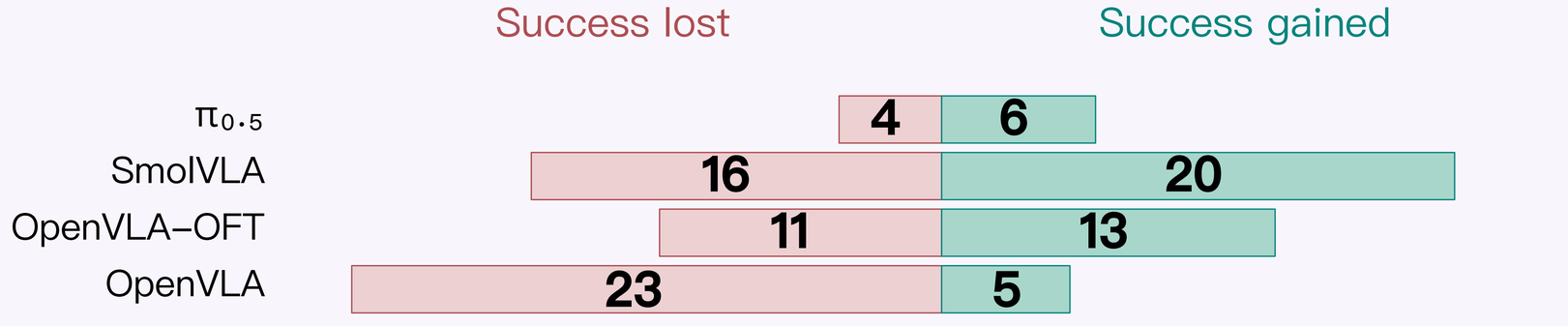
40 tasks × 12 starts = 480 pairs



44/480 outcomes differ · 437 → 435 successes · schematic widths



SmolVLA
100 pairs per system



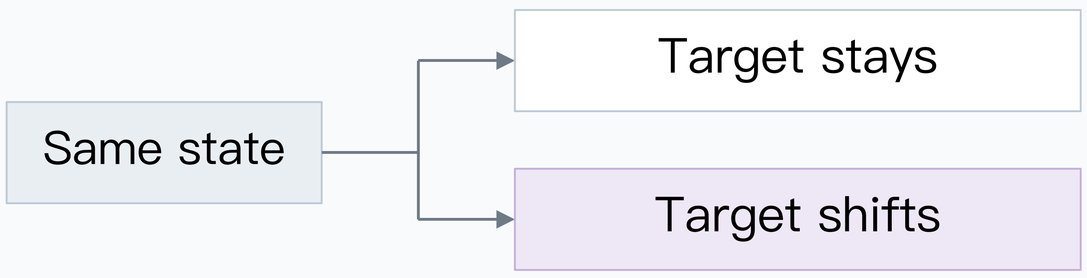
Total successes
65 → 69

Success lost
16

Success gained
20

Higher success rates can hide new failures.

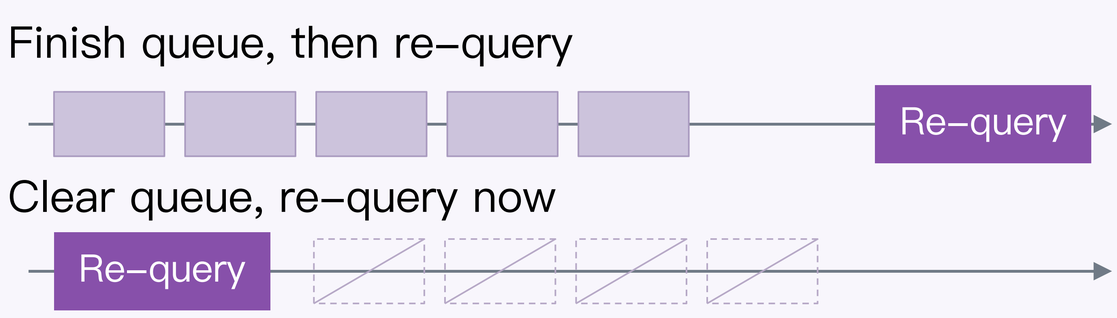
Does success mean correction?



System	After shift	Both conditions	Shift only
SmolVLA	**21/50**	21	**0**
OpenVLA–OFT	**23/50**	15	**8**

Similar totals hide different success compositions.
Success after a shift does not prove correction.

Does execution change the result?



System	Finish queue	Re–query	Gain [95% CI]
$\pi_{0.5}$	40/50	48/50	**+16 pp [4, 28]**
SmolVLA	21/50	36/50	**+30 pp [12, 48]**

Same checkpoints, 50 matched cases per system.
Outcomes depend on execution settings.

Report matched outcomes, not just a success rate.

State what changed, which cases were matched, and how the policy was executed.